MAD Paper Solution

Q1. Short Questions (1 Mark Each × 3 = 3 Marks)

1. What is the role of Canvas in Android development?
Canvas in Android provides a surface on which developers can draw graphics such as shapes, text, images, and custom UI elements using methods like drawLine(), drawRect(), drawText().

2. What is menu?
A menu in Android is a UI component that provides a list of options/actions for the user, usually displayed in the app bar or overflow menu.

3. Which cellular network type is based on 4G technology?
The cellular network based on 4G technology is **LTE (Long Term Evolution)**.

Q2. Objective Type / MCQs / T-F / Fill in the blanks (1 Mark Each × 7 = 7 Marks)

1. Which line of code starts Activity2 from Activity1?
✅ (C) Intent intent = new Intent(Activity1.this, Activity2.class); startActivity(intent);

2. Attribute to set activity to landscape orientation:
✅ (D) android:screenOrientation="landscape"

3. Configuration file for application resides in Res folder and contains java code. (T/F)
❌ False – Java code is kept in src/ folder, not in res/.

4. Android is an open source and Window-based Operating System. (T/F)
❌ False – Android is open source and **Linux-based**, not Windows-based.

5. Tool used to create certificates for signing Android apps:
✅ (C) keytool

6. AVD stands for?
✅ **Android Virtual Device**

7. Pending Intent in Android means:

✅ (C) It will fire at a future point of time.

Q3. Two Questions (2 Marks Each = 4 Marks)

1. Explain use of manifest file in Android.

- The AndroidManifest.xml file provides essential information about the app to the Android system.
- It declares app components (activities, services, broadcast receivers, content providers), permissions, minimum SDK version, and app metadata.

2. Write a short note on packaging and signing an Android app.

- Packaging: Android apps are packaged into an .apk file containing code, resources, assets, and manifest.
- Signing: Before installation, an app must be digitally signed with a developer key to ensure authenticity and integrity.

Q4. Two Questions (3 Marks Each = 6 Marks)

1. Explain implicit intent with example.

- Implicit intents do not specify the target component but declare an action to be performed.
- The Android system resolves it to an appropriate app.
- Example:

```
Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse("http://www.google.com"));

startActivity(intent);
```

This opens the web browser to the given URL.

2. Explain Android Layout with example.

- Layout defines the structure of UI in an Android app (arrangement of elements).
- Types: LinearLayout, RelativeLayout, ConstraintLayout, FrameLayout, etc.

- Example (XML):

```
<LinearLayout android:orientation="vertical"
   android:layout_width="match_parent"
   android:layout_height="match_parent">
   <TextView android:text="Hello World"/>
   <Button android:text="Click Me"/>
</LinearLayout>
```

Q5. Attempt Any TWO (5 Marks Each = 10 Marks)

1. List User Interface elements and explain any 3.

- UI elements: TextView, EditText, Button, ImageView, CheckBox, RadioButton, ProgressBar, Spinner, etc.
- Explanation (any 3):
  - **TextView**: Displays static text to the user.
  - **Button**: A clickable element used to perform an action.
  - **EditText**: Allows user to enter text input.

OR

2. Draw and explain Android architecture.

- **Application Layer**: Installed apps like SMS, Browser.
- **Application Framework**: Provides APIs (Activity Manager, Window Manager).
- **Libraries & Android Runtime**: Includes SQLite, OpenGL, and Dalvik/ART virtual machine.
- **Linux Kernel**: Handles hardware drivers, memory, and power management.

OR

3. Define cellular networks and their use in mobile computing.

- Cellular networks are wireless networks divided into cells, each served by a base station.
- Used in mobile computing for voice, SMS, and internet services.
- Example: 3G, 4G (LTE), 5G.

Q6. Long Questions (5 Marks Each × 2 = 10 Marks)

1. Content Provider Scenario
(a) **Purpose of Content Provider:**

- Content Provider in Android enables sharing of data between applications securely.
- It acts as an interface to access structured data (e.g., student notes, assignments).

(b) **Advantages:**

1. Provides controlled access to data (with permissions).
2. Enables inter-app communication without exposing databases directly.

(c) **Limitation:**

- Complex to implement and may reduce performance compared to direct access.

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2. Describe different types of resources in Android and how they are managed.

- **Drawable resources:** Images, shapes, icons.
- **Layout resources:** XML files defining UI structure.
- **String resources:** Text used in the app.
- **Color & Style resources:** Theme colors, styles for consistency.
- **Animation resources:** Define view animations.

Management: Resources are placed in the res/ folder and accessed using R class (e.g., R.layout.main_activity). Android automatically manages resources based on device configuration (screen size, language, orientation).

OR

What is Android SDK and its components?

- **Android SDK (Software Development Kit):** Set of tools to develop Android apps.

- **Components:**
    1. **SDK Tools:** Debugging, testing, building.
    2. **SDK Platform Tools:** adb, fastboot.
    3. **Android Emulator:** Testing apps without device.
    4. **Libraries:** APIs for Android features.
    5. **Build Tools:** Compilers, packaging tools.